3.5.1 Radioactivity

- ### Evidence for the nucleus

  Qualitative study of Rutherford scattering.

- ### α, β and γ radiation

  Their properties and experimental identification using simple absorption experiments; applications e.g. to relative hazards of exposure to humans.

  The inverse square law for γ radiation, $I = \dfrac{k}{x^2}$

  including its experimental verification; applications, e.g. to safe handling of radioactive sources.

  Background radiation; examples of its origins and experimental elimination from calculations.

- ### Radioactive decay

  Random nature of radioactive decay; constant decay probability of a given nucleus;

  $\dfrac{\Delta N}{\Delta t} = -\lambda N, \quad N = N_0 e^{-\lambda t}$

  Use of activity $A = \lambda N$

  Half life, $T_{1/2} = \dfrac{\ln 2}{\lambda}$; determination from

  graphical decay data including decay curves and log graphs; applications e.g. relevance to storage of radioactive waste, radioactive dating.

- ### Nuclear instability

  Graph of N against Z for stable nuclei.

  Possible decay modes of unstable nuclei including α, β^+, β^- and electron capture.

  Changes of N and Z caused by radioactive decay and representation in simple decay equations.

  Existence of nuclear excited states; γ ray emission; application e.g. use of technetium-99m as a γ source in medical diagnosis.

- ## Nuclear radius

Estimate of radius from closest approach of alpha particles and determination of radius from electron diffraction; knowledge of typical values.

Dependence of radius on nucleon number

$R = r_0 A^{1/3}$

derived from experimental data.

Calculation of nuclear density.